

Comparative Analysis of Machine Learning Algorithms for Enhanced Diabetes Prediction

Mini-project-I Report

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For the degree of B.Tech

in

Computer Science and Engineering (AIML)

by

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Certificate

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“Comparative Analysis of Machine Learning Classifiers and Feature Selection Techniques for Enhanced Diabetes Prediction”

is a bonafide work done by

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Mini-Project Approval for B.Tech

This is to certify that the dissertation entitled “**Comparative Analysis of Machine Learning Classifiers and Feature Selection Techniques for Enhanced Diabetes Prediction**” is a bonafide work done by **JADHAV AKHILESH NIVRUTTI, JAIN VIDHAN AMIT, SADADEKAR ISHITA DEEPAK, and LOKHANDE OMKAR SUBHASH** under the supervision of **Dr. Saguna Ingle**. This dissertation has been approved for the award of Mini Project in Computer Science and Engineering (AIML), D. Y. Patil Deemed to be University.

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Abstract

Diabetes Mellitus is a great burden on public health due to its association with increased risks of heart disease, kidney problems, nerve damage, and unnecessary complications. While laboratory tests are a must for diagnosis, prediction models built on machine learning can help identify patients at higher risk by detecting relevant trends in the values of routine clinical attributes. This paper describes a comparative assessment of five machine learning classification algorithms—Decision Tree, K-Nearest Neighbors (KNN), Logistic Regression, Random Forest, and Support Vector Machine (SVM)—coupled with six feature selection approaches: ANOVA K-Best, Mutual Information K-Best, Recursive Feature Elimination with Logistic Regression, L1-regularized Logistic Regression, Random Forest importance, and Extra Trees importance. In total, thirty pipelines were evaluated in terms of accuracy, precision, recall, F1-score, and ROC-AUC.

The exploratory analysis and iterative feature selection consistently showed the dominance of *HbA1c level*, *blood glucose level*, *age*, *BMI*, and *hypertension* as the main attributes predictive of diabetes. Results of the experiment indicated the trade-off among the models based on certain metrics. **RFE_LogReg + KNN** reached the highest test accuracy (0.9696), precision (0.9588), and F1-score (0.7896) and thus became the best threshold-based classifier. Conversely, **L1_LogReg + SVM** was found to deliver the greatest ROC-AUC (0.9700) along with the highest recall (0.9224) and hence the best class-separability. Random Forest proved to be the most well-rounded model according to metrics, whereas Decision Tree demonstrated high recall with extremely high false positives.

This study proves that selection of a machine learning classifier suitable for diabetes prediction is not based solely on accuracy metrics since class imbalance plays a critical part. Depending on the specific goal of clinical practice, different classifiers are to be used.

Keywords: Diabetes prediction, machine learning, feature selection, supervised classification, ROC-AUC, screening analytics

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Chapter 1

Introduction

Today's healthcare systems have access to many structured and semi-structured data types, ranging from patient demographics information to their clinical records and biochemical parameters. In turn, machine learning can help discover meaningful patterns within collected data and utilize them to develop prediction models capable of suggesting possible medical interventions. The disease chosen for analysis within this project is Diabetes Mellitus (DM). It is known to be a chronic metabolic condition. Failure to diagnose it at an early stage leads to severe medical conditions related to the heart, kidneys, eyes, nervous system, and blood vessels. There is an array of established diagnostic procedures aimed at identifying DM. These include fasting plasma glucose tests, oral glucose tolerance tests, and determination of the hemoglobin A1c (HbA1c). Such diagnostic techniques are well-recognized within the medical community. However, they are usually employed after initial clinical observations reveal certain symptoms. Predictive modeling can serve as a supplementary procedure that facilitates the early identification of patients' predisposition to the disease and the need for further testing. On the other hand, medical datasets tend to contain features that add little to the prediction process and negatively affect modeling. The present research focuses on structured experimental evaluation of the application of multiple feature selection approaches together with various supervised machine learning algorithms in predicting diabetes. The primary aim of this research is not to determine the algorithm and feature subset combination resulting in the highest predictive accuracy but rather to assess the impact of applying a particular classifier on a specific feature set in terms of different performance metrics. In this regard, special emphasis is placed on recall and ROC-AUC.

1.1 Objectives

The objectives of the study are as follows:

1. Comparison of 30 machine learning pipelines generated by combining 6 feature selection approaches with 5 classifiers.

2. Performing thorough exploratory analysis and examining the associations between correlation matrices, feature selection results, and their importance.
3. Model performance assessment through multiple metrics to prove that the choice of the best performing classifier is related to its clinical application.
4. Confusion matrix analysis particularly of models where recall is more important.

1.2 Literature Review and Taxonomy

Various machine learning approaches for predicting the probability of diabetes based on clinical parameters and lifestyle information have been extensively investigated in the literature related to the field of health care analytics. Among these approaches, Naive Bayes, Decision Tree, Logistic Regression, Random Forest, Support Vector Machines, and Neural Network have been employed to predict the onset of diabetes using the given clinical and lifestyle data. While many studies report successful results, the number of papers reporting the overall performance assessment of their proposed algorithm through various measures, including some healthcare-related ones such as sensitivity or recall, is small. From a general perspective, machine learning algorithms can be categorized into three categories according to their approach, namely supervised learning, unsupervised learning, and semi-supervised learning. The task of predicting the likelihood of diabetes from the input data is a supervised learning problem because the training process requires not only the input features but also the corresponding output labels to learn the mapping. For this task specifically, the goal of the supervised learning method is to discover the link between patient's attributes, such as Glucose and HbA1c levels, and the outcome class – having diabetes or not. It is clear that choosing relevant features is also a crucial part of diabetes prediction. Among feature selection algorithms, filter algorithms are represented by ANOVA and Mutual Information; both evaluate each feature independently considering its relation with the target attribute. Embedded feature selection algorithms include L1-regularized logistic regression and tree-based feature importance; they choose features while constructing the classifier model. Finally, wrapper methods involve iterative elimination procedures and are represented by Recursive Feature Elimination (RFE). The contribution of this paper is that the issue of feature selection is explicitly addressed as an essential part of the comparative analysis, rather than being considered as a pre-processing stage of the machine learning problem. Instead of applying each classification method to only one selected subset of features, as in the previous papers, six different feature selection methods are employed in this study on all classifiers.

1.3 Motivation: The Mathematics of Clinical Error

In the case of classification in healthcare, a model can demonstrate high accuracy for reasons not related to the effectiveness of the prediction itself. One of those reasons is the dominance of one of the classes among all others, which will inevitably cause high accuracy for the algorithm. In diabetes prediction, high accuracy does not always mean that the model will be effective, since the false-negative rate can be higher than acceptable due to the predominance of one class over another.

The two most critical types of errors are:

1. **False Negative (FN):** The patient is diabetic, but the model identifies the patient as healthy.
2. **False Positive (FP):** The patient is healthy, but the model predicts that he/she has diabetes.

Even though these two errors can be equally detrimental to the work of the model in general, the consequences of false-negative errors might be much more severe because patients' lives could be threatened if a false prediction is not detected at the right time. That is why this paper focuses on recall and ROC-AUC score.

Recall (Sensitivity) is a metric that indicates how many diabetic patients are detected in total. Therefore, recall is especially significant in the case of diabetes screening because high sensitivity means that the model can recognize all diabetic patients. However, low sensitivity is acceptable in the case of precision-based prediction.

Chapter 2

Proposed Methodology

2.1 Exploratory Data Analysis

The first step of the methodology was to understand the overall structure and characteristics of the dataset. By examining the distribution plots, it was observed that the dataset contains both continuous biomedical features and encoded demographic variables. Another important finding at this stage was that the target variable was imbalanced, with non-diabetic cases being much higher than diabetic cases. Because of this imbalance, it was necessary to evaluate the models using more than just accuracy.

The correlation heatmap shown in Figure 2.1 indicates that diabetes has the strongest positive relationship with blood glucose level (around 0.42) and HbA1c level (around 0.40). These are followed by age (around 0.26), BMI (around 0.21), and hypertension (around 0.20). Although these correlation values are moderate rather than extremely high, they suggest that diabetes cannot be explained by a single factor alone. Instead, it is influenced by a combination of clinical and demographic variables. At the same time, the repeated appearance of the same important features in the correlation analysis gives a meaningful foundation for the feature selection results discussed later.

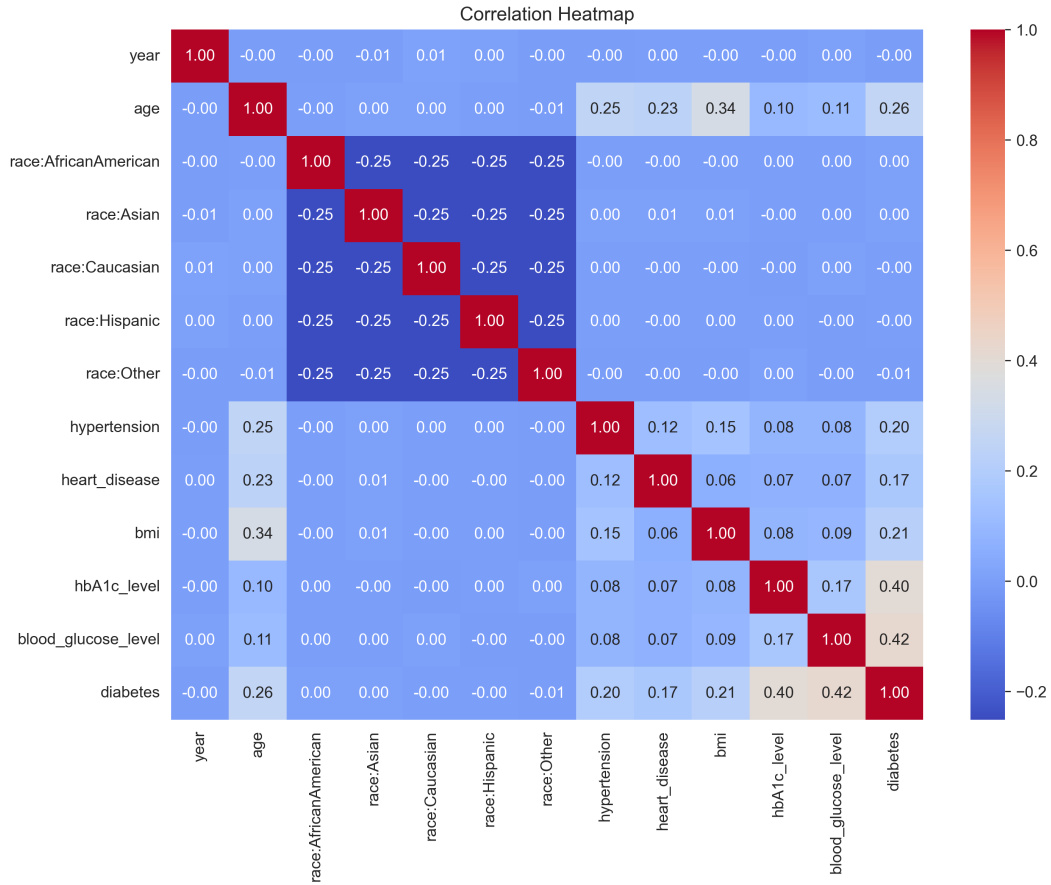


Figure 2.1: Correlation heatmap showing that the target variable has the strongest positive relationship with blood glucose level, HbA1c level, age, BMI, and hypertension.

Figure 2.2 gives a clearer view of how the individual features are distributed in the dataset. Continuous variables such as BMI, HbA1c, and blood glucose level show a wide range of values, while encoded binary features are mostly concentrated at 0 and 1. The target variable is also noticeably imbalanced, with non-diabetic cases appearing much more frequently than diabetic cases. This imbalance helps explain why a model may achieve high overall accuracy while still performing poorly in identifying the minority diabetic class, especially in terms of precision or recall.

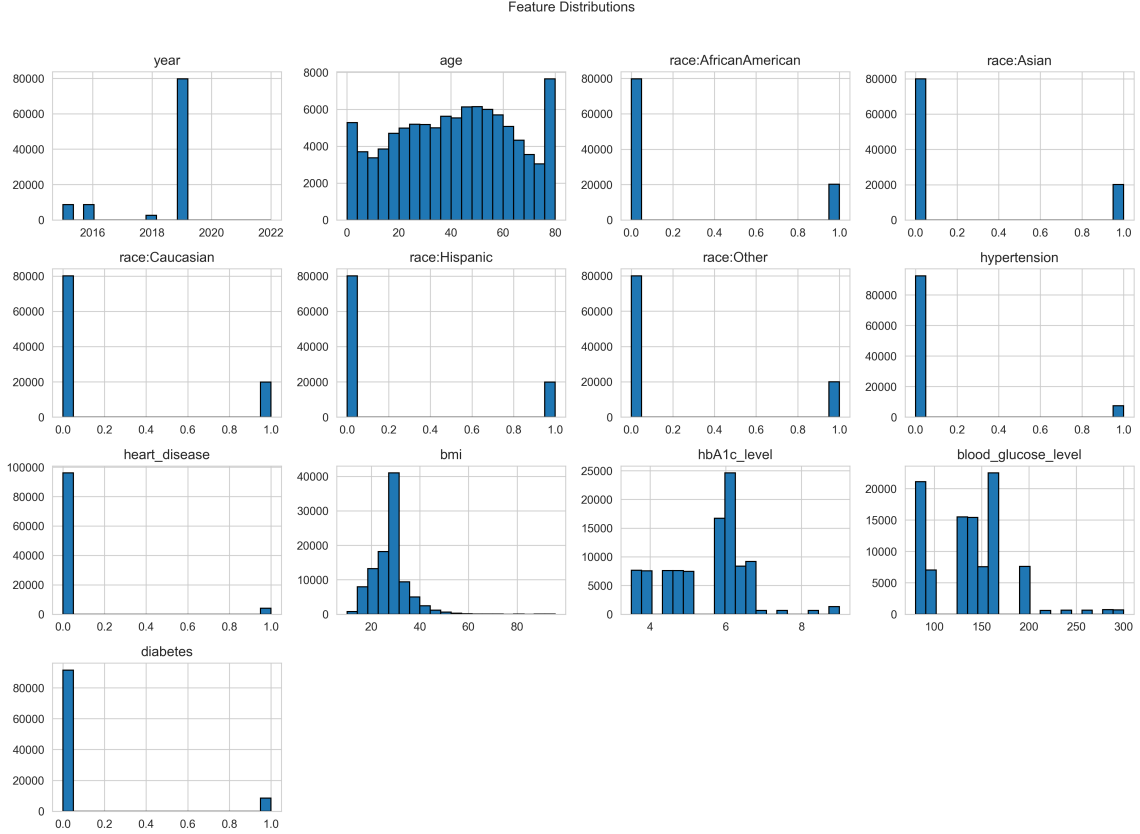


Figure 2.2: Distribution plots of the major input features and the diabetes target variable. The imbalance in the target class supports the use of additional evaluation metrics such as recall, F1-score, and ROC-AUC instead of depending only on accuracy.

2.2 Feature Selection

In this study, six different feature selection methods were used to identify the most relevant variables for diabetes prediction:

1. **ANOVA K-Best:** Selects features that show the strongest linear relationship with the target class using the F-statistic.
2. **Mutual Information K-Best:** Measures how much information each feature provides about the target variable.
3. **L1-regularized Logistic Regression:** Performs feature selection during training by reducing weaker feature coefficients toward zero.
4. **Recursive Feature Elimination (RFE) with Logistic Regression:** Repeatedly removes less important features to obtain a smaller and more informative subset.
5. **Random Forest Importance:** Ranks features based on how much they contribute to reducing impurity across the trees.

6. **Extra Trees Importance:** Similar to Random Forest importance, but uses a more randomized tree-based approach to estimate feature importance.

One of the important findings of this study is that these different feature selection methods produced highly consistent results. As shown in Figure 2.3, blood glucose level, HbA1c level, and age were selected by all six methods. In addition, BMI and hypertension were also chosen repeatedly by most of the selectors. This pattern suggests that the main predictive information is concentrated in a small set of clinically meaningful features.

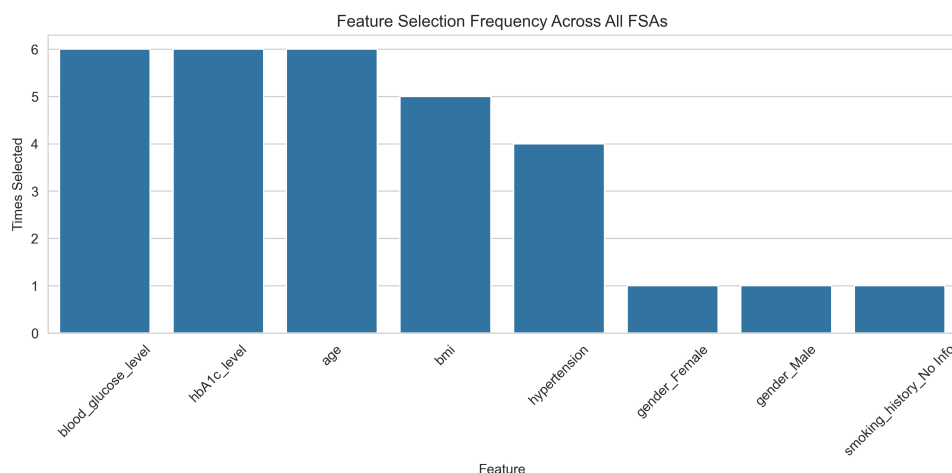
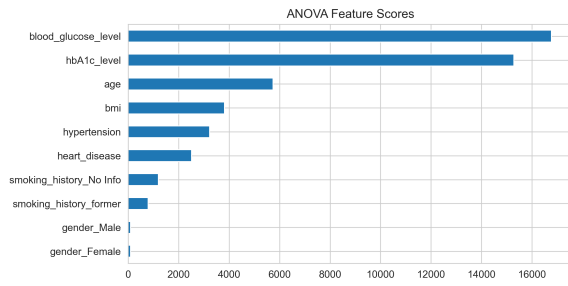
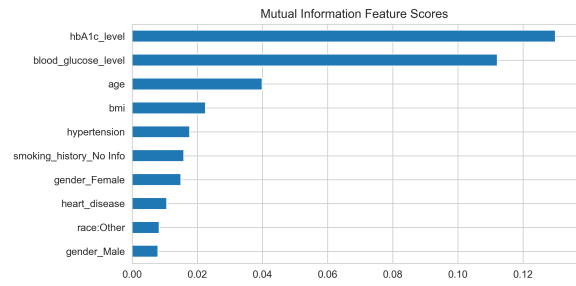


Figure 2.3: Frequency of feature selection across all six methods. HbA1c level, blood glucose level, and age were selected in every case, while BMI and hypertension were also retained consistently.

The individual feature importance plots also support the same conclusion. ANOVA ranked blood glucose level and HbA1c level much higher than the remaining features. Mutual Information, Random Forest importance, Extra Trees importance, and L1-based selection also highlighted the same core metabolic biomarkers as the strongest predictors. Some methods additionally selected secondary demographic variables such as gender, smoking history, race, year, or location-related dummy variables. These features may provide some extra predictive value, but their contribution is still much smaller compared to the glucose-related clinical measurements.

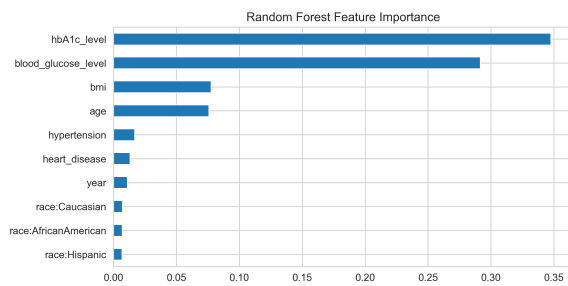


(a) ANOVA feature scores

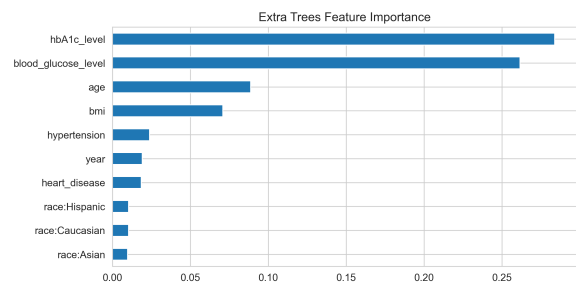


(b) Mutual Information scores

Figure 2.4: The filter-based feature selection methods clearly show that blood glucose level and HbA1c level are the most important predictors.



(a) Random Forest importance



(b) Extra Trees importance

Figure 2.5: Tree-based feature importance results from Random Forest and Extra Trees again highlight HbA1c, blood glucose, age, and BMI as the main predictive variables.

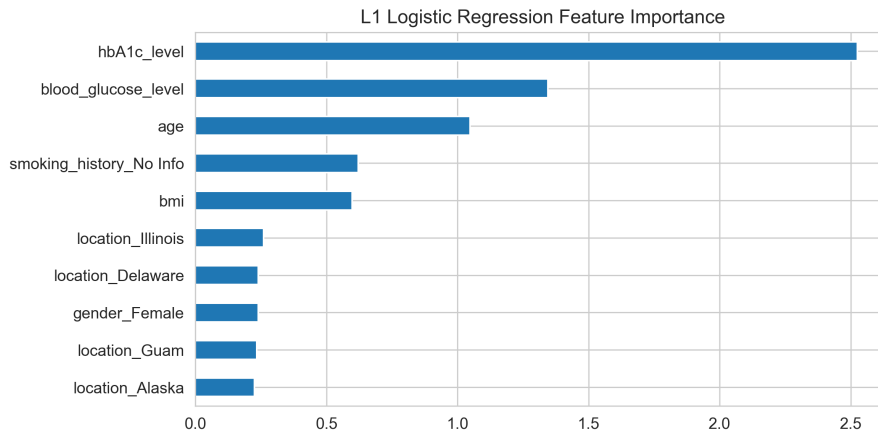


Figure 2.6: The embedded feature importance obtained from L1-regularized Logistic Regression shows that the most significant retained features include HbA1c, blood glucose, age, smoking history, BMI, and a few sparse encoded indicators related to location and gender.

2.3 Classifier Matrix

This study uses five supervised machine learning classifiers: Decision Tree, K-Nearest Neighbors (KNN), Logistic Regression, Random Forest, and Support Vector Machine (SVM). Each of these classifiers was trained using the feature subsets produced by the six different feature selection methods. As a result, a total of 30 classifier-feature selection combinations were created and evaluated.

2.4 Evaluation Metrics

The performance of the models was measured using standard evaluation metrics obtained from the confusion matrix:

- **Accuracy** measures the overall proportion of correctly classified samples.
- **Precision** indicates how many of the samples predicted as diabetic were actually diabetic.
- **Recall** shows how many of the actual diabetic cases were correctly identified by the model.
- **F1-score** provides a balance between precision and recall.

Mathematically, these metrics are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2.1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2.2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.3)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2.4)$$

In addition to these metrics, ROC-AUC analysis was carried out to assess the discrimination capacity of the models for the separation of the two classes by varying thresholds. This particular approach is particularly relevant in this case, given that the data is unbalanced, and provides a measure of discrimination power, in addition to classification performance

Chapter 3

Comprehensive Results and Algorithmic Analysis

All five classifiers were crossed with all six selectors, resulting in thirty evaluated pipelines. Table 3.1 presents the complete test-set results.

Table 3.1: Master results table: performance of all 30 classifier-selector pipelines

Feature Selector	Classifier	Accuracy	Precision	Recall	F1	ROC-AUC
ExtraTrees_Importance	DecisionTree	0.7819	0.2715	0.9306	0.4204	0.9418
ANOVA_KBest	DecisionTree	0.7819	0.2715	0.9306	0.4204	0.9418
L1_LogReg	DecisionTree	0.7819	0.2715	0.9306	0.4204	0.9418
MutualInfo_KBest	DecisionTree	0.7819	0.2715	0.9306	0.4204	0.9418
RFE_LogReg	DecisionTree	0.7819	0.2715	0.9306	0.4204	0.9418
RF_Importance	DecisionTree	0.7819	0.2715	0.9306	0.4204	0.9418
MutualInfo_KBest	KNN	0.9683	0.9449	0.6653	0.7808	0.9209
ANOVA_KBest	KNN	0.9683	0.9449	0.6653	0.7808	0.9209
ExtraTrees_Importance	KNN	0.9683	0.9449	0.6653	0.7808	0.9209
RF_Importance	KNN	0.9683	0.9449	0.6653	0.7808	0.9209
L1_LogReg	KNN	0.9666	0.9372	0.6500	0.7676	0.9149
RFE_LogReg	KNN	0.9696	0.9588	0.6712	0.7896	0.9136
MutualInfo_KBest	LogisticRegression	0.8844	0.4150	0.8771	0.5634	0.9600
ANOVA_KBest	LogisticRegression	0.8844	0.4150	0.8771	0.5634	0.9600
RF_Importance	LogisticRegression	0.8844	0.4150	0.8771	0.5634	0.9600
ExtraTrees_Importance	LogisticRegression	0.8844	0.4150	0.8771	0.5634	0.9600
L1_LogReg	LogisticRegression	0.8809	0.4071	0.8788	0.5564	0.9592
RFE_LogReg	LogisticRegression	0.8694	0.3819	0.8676	0.5304	0.9522
L1_LogReg	RandomForest	0.9669	0.8886	0.6988	0.7824	0.9492
ANOVA_KBest	RandomForest	0.9647	0.8617	0.6965	0.7703	0.9488
MutualInfo_KBest	RandomForest	0.9649	0.8617	0.7000	0.7725	0.9482
ExtraTrees_Importance	RandomForest	0.9649	0.8617	0.7000	0.7725	0.9482
RF_Importance	RandomForest	0.9651	0.8646	0.6988	0.7729	0.9480
RFE_LogReg	RandomForest	0.9311	0.5696	0.7747	0.6565	0.9215
L1_LogReg	SVM	0.8791	0.4070	0.9224	0.5647	0.9700
ANOVA_KBest	SVM	0.8818	0.4127	0.9229	0.5703	0.9692
RF_Importance	SVM	0.8818	0.4127	0.9229	0.5703	0.9692
MutualInfo_KBest	SVM	0.8818	0.4127	0.9229	0.5703	0.9692
ExtraTrees_Importance	SVM	0.8818	0.4127	0.9229	0.5703	0.9692
RFE_LogReg	SVM	0.8577	0.3659	0.9194	0.5234	0.9655

3.1 Heatmap-Level Interpretation

Based on the accuracy heatmap presented in Figure 3.1, the KNN algorithm demonstrates the best performance in terms of the threshold-based accuracy criterion. The accuracy of KNN classifiers is fairly close to 0.968 and attains the maximal value of 0.9696 in combination with RFE selector. The performance of Random Forest classifiers is also high and close to 0.965 using several feature selection approaches. Logistic Regression and SVM algorithms are characterized by lower accuracy scores that are around 0.88, while the performance of Decision Tree does not change much, being close to 0.782 with all the selectors applied. However, Figure 3.2 presents a different picture in terms of the ROC-AUC criterion. There, SVM is clearly shown to be the best-performing algorithm. Its maximal ROC-AUC value is 0.9700 with L1-regularized feature selection. The other best values obtained with SVM are around 0.969 for the other feature selectors. The second-best classifier is Logistic Regression in terms of ROC-AUC scores. At the same time, KNN models that perform excellently for direct classification do not demonstrate the best performance in terms of ROC-AUC and are characterized by ROC-AUC values between 0.914 and 0.921. Thus, this discrepancy is one of the main insights provided by the present analysis

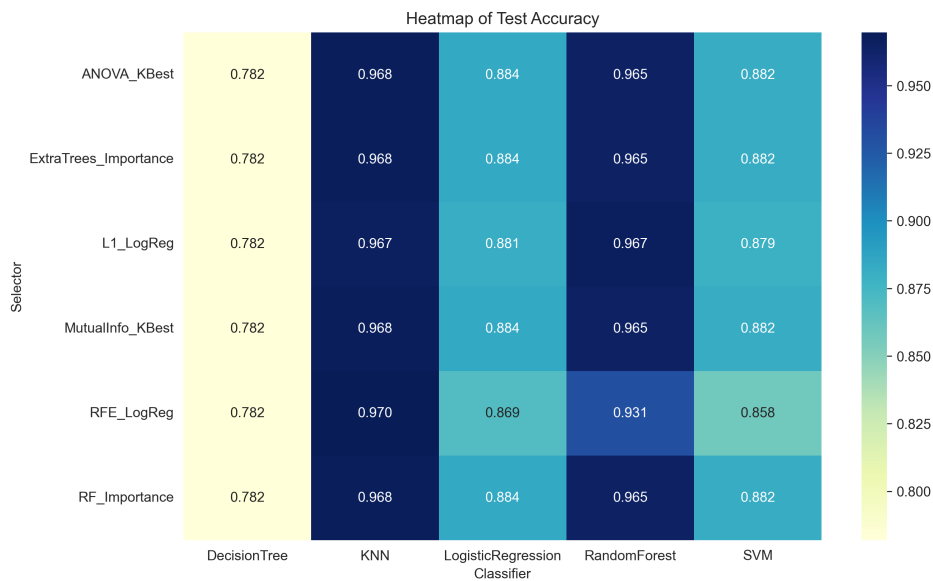


Figure 3.1: Accuracy heatmap presenting the best-performing classifier-selector combinations

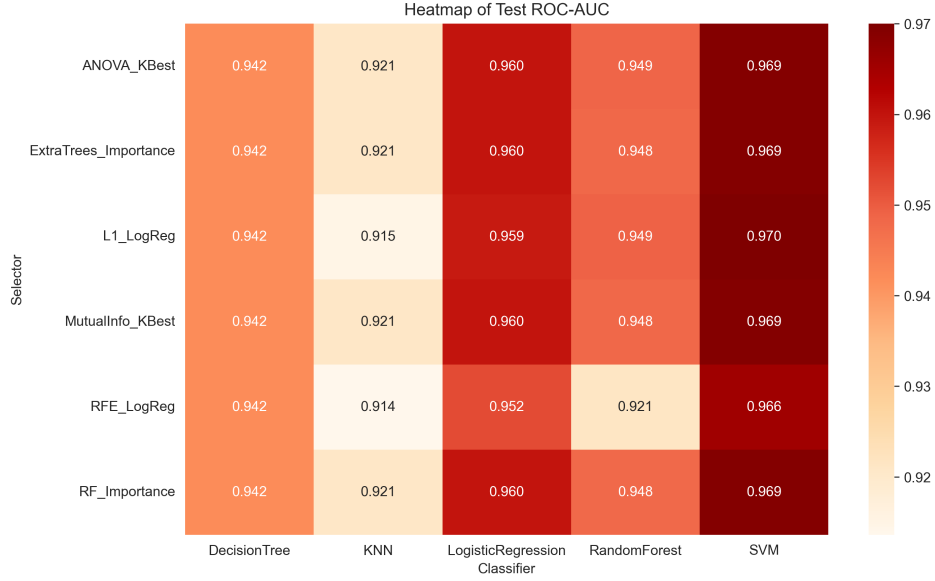


Figure 3.2: ROC-AUC heatmap presenting the best-performing classifier-selector combinations

3.2 Best Pipelines by Metric

It should be noted that the optimal pipeline will depend on the chosen criterion for assessing its quality:

- **Accuracy of testing:** $\text{RFE_LogReg} + \text{KNN} = 0.9696$
- **Precision of testing:** $\text{RFE_LogReg} + \text{KNN} = 0.9588$
- **F1-score of testing:** $\text{RFE_LogReg} + \text{KNN} = 0.7896$
- **Recall of testing:** Decision Tree with all selectors = 0.9306
- **ROC-AUC of testing:** $\text{L1_LogReg} + \text{SVM} = 0.9700$

Thus, it can be seen that the phrase “*the best model*” does not make sense without specifying the assessment criterion. The recall and ROC-AUC criteria are important for screening purposes, making the SVM pipeline more suitable. If minimizing false positives is a priority in a classification problem, then KNN would be the better choice. Random Forest falls somewhere between these two options.

3.3 Classifier-Specific Analysis

Decision Tree. From all models, Decision Tree provided the highest value of recall — 0.9306. At the same time, its precision was very low (0.2715), while its F1-score

was equal to 0.4204. This demonstrates that the model makes numerous predictions of patients being diabetic and can detect virtually all real positive cases. Nevertheless, it makes numerous false predictions of patients, which makes it applicable solely for a general screening stage only.

K-Nearest Neighbors. KNN is the most powerful model from the perspective of threshold classification performance. In particular, the model combined with the use of RFE selector has achieved the highest accuracy (0.9696), precision (0.9588), and F1-score (0.7896). Thus, if KNN predicts a patient being diabetic, the probability of this prediction being correct is quite high. Still, KNN has the lowest recall among all models considered in this study (best result of 0.6712). Hence, KNN is applicable more to confident classification purposes than medical screening.

Logistic Regression. The model has demonstrated solid performance, especially on ROC-AUC and recall indicators. In particular, Logistic Regression showed consistent results in ANOVA, Mutual Information, Random Forest importance, and Extra Trees importance selectors. At the same time, its precision has remained relatively low (around 0.41), making Logistic Regression resemble a screening-oriented model rather than a high-precision classifier.

Random Forest. Random Forest provided the best-balanced results among all models in terms of different metrics. Specifically, the model in combination with L1 feature selector showed accuracy of 0.9669, precision of 0.8886, recall of 0.6988, F1-score of 0.7824, and ROC-AUC of 0.9492. Despite the fact that Random Forest cannot boast of the best indicator on any measure, its performance remains quite good on each metric without any serious drawbacks.

Support Vector Machine. SVM has provided the highest ROC-AUC values and very high recall scores. Best configuration of SVM is L1 with the score of 0.9700 on ROC-AUC and 0.9224 on recall measures. Therefore, SVM proves to be a powerful separator between two classes and is capable of making an accurate diagnosis. Unfortunately, precision of the model is relatively low (0.4070), so SVM requires some further confirmation tests.

3.4 Top-5 ROC Curves

Figure 3.3 shows the ROC curves of the top five pipeline combinations. All the five pipelines mentioned above have been developed using SVM, with the only difference being the feature selection algorithm used. This is an additional proof that SVM always proves to be the best model when evaluating class separation independently of any threshold. L1-SVM appears to be slightly superior to the rest, but all five curves are almost identical to one another. It indicates that, once important markers are kept, SVM does not depend

on the type of feature selection.

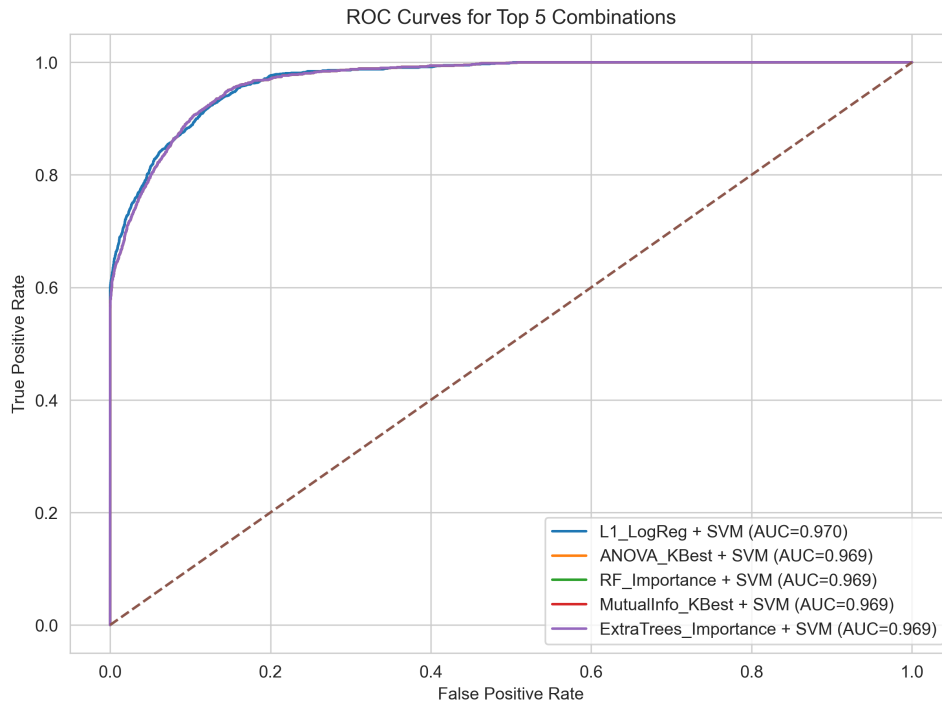


Figure 3.3: ROC curves of the top five pipeline combinations.

3.5 Confusion Matrix of the Best ROC-AUC Model

The confusion matrix of L1 LogReg + SVM classifier is illustrated by Figure 3.4, whereas the key characteristics of the algorithm are presented in Table 3.2. The classifier successfully detects 1568 cases with diabetes and makes only 132 false negatives, and that explains the high level of recall. However, the algorithm makes 2285 false positives, and that reduces its precision.

Such a classifier has a practical meaning for clinical diagnostics, as the number of false negatives is minimal, and thus the model is highly effective in use for screening patients for diabetes. Still, the number of false positives is quite high, and thus it is not advisable to use the model as a basis for decision-making. It is much better to use this model as a first-step screening test, and then perform additional checks to make sure the diagnosis was made correctly.

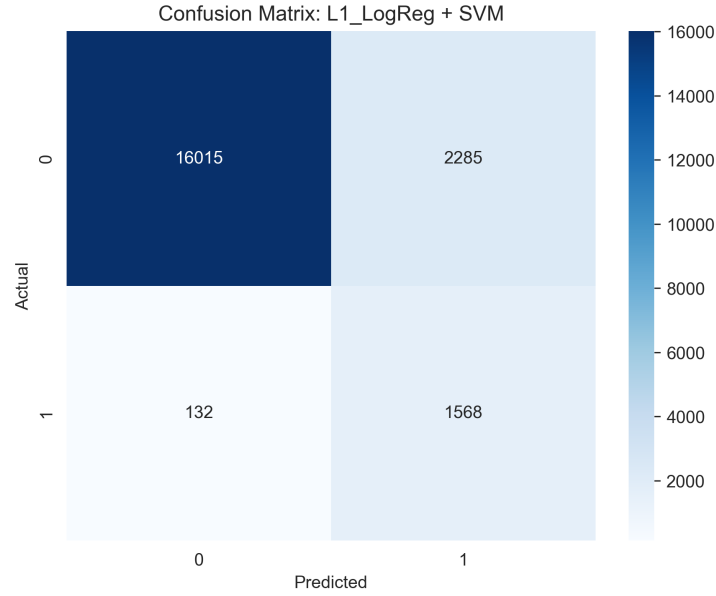


Figure 3.4: Confusion matrix of the selected model of L1 LogReg + SVM classifier. This classifier makes very few false negatives, but also many false positives.

Table 3.2: Confusion matrix values for L1 LogReg + SVM

	Predicted 0	Predicted 1
Actual 0	16015	2285
Actual 1	132	1568

3.6 Average Performance by Classifier

Averaging each metric across all selectors provides a broader view of classifier behavior, shown in Table 3.3.

Table 3.3: Average test performance across all feature selection algorithms

Classifier	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	0.7819	0.2715	0.9306	0.4204	0.9418
KNN	0.9682	0.9459	0.6637	0.7801	0.9187
Logistic Regression	0.8813	0.4082	0.8758	0.5567	0.9586
Random Forest	0.9596	0.8180	0.7115	0.7545	0.9440
SVM	0.8773	0.4040	0.9222	0.5616	0.9687

Chapter 4

Discussion

Several conclusions can be drawn from the experimental results of this study.

The first notable observation relates to the dominant position of only a few predictors in the dataset. In both correlation heatmap, ANOVA scores, Mutual Information scores, tree-based feature importance, and L1 embedded selection, there were always several predictors that stood out from the rest: *HbA1c level*, *blood glucose level*, *age*, *BMI*, and *hypertension*. The significance of this finding stems from the independence of feature selection method: no matter whether filter, wrapper, or embedded approach was used, the main predictive signals always remained the same.

Another important conclusion relates to the impact of class distribution imbalance on evaluation and interpretation of the model performance. It is possible that a model can demonstrate an impressive value of certain metrics (such as accuracy) while failing to satisfy the clinical requirements. Because of this reason, the current study evaluates model performance through multiple criteria, including both threshold-based (accuracy, precision) and threshold-independent (ROC-AUC). As an example, KNN demonstrates excellent accuracy and precision values, but its ability to detect diabetic patients is weaker compared to SVM, which performs much better in terms of recall and ROC-AUC, making it useful in predicting the disease despite higher numbers of false positives.

Another important finding is related to the need for considering distinct use cases when selecting a model for diabetes detection:

- For **broad screening**, when missing a true diabetic patient is unacceptable, SVM is the preferable choice due to its superior recall and ROC-AUC performance.
- For **high-confidence positive prediction**, KNN will do better due to its superior precision values.
- For a **balanced clinical decision-support system**, Random Forest is the most practical option as it is less biased towards any single metric.

Finally, the relatively modest difference in performance between several feature selection

methods indicates that the majority of predictive signals is conveyed by a relatively narrow range of important factors. As long as the metabolic biomarkers have been identified and used for modeling, the classifier behavior starts playing a bigger role in predicting outcomes. This trend is observable in ROC curves, where SVM-based classification pipelines demonstrate similar performance.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

In this research, a comparison between different machine learning models and their performances using different combinations of feature selection was conducted for the purpose of predicting the occurrence of diabetes. Thirty models based on six feature selection techniques and five supervised machine learning classifiers were tested to find the most optimal configuration and predict diabetes in patients. It was determined that the most consistent predictors of diabetes in this research include HbA1c level, blood glucose level, age, BMI, and hypertension. These indicators were repeated frequently in exploratory data analysis, correlation results, and various feature selection methods.

It can be concluded from the findings above that there is no universally perfect machine learning algorithm to use for threshold-based and other types of prediction models. The RFE LogReg + KNN pair showed the best accuracy (0.9696), precision (0.9588), and F1-score (0.7896). On the other hand, the L1 LogReg + SVM algorithm had the highest ROC-AUC (0.9700) and very high recall score (0.9224), thus making it better suited for screening purposes. Lastly, Random Forest showed balanced scores in all key performance indicators whereas Decision Tree classifier scored very well in recall but generated numerous false positives.

Therefore, the primary finding is that model selection should be based on clinical goals. To decrease the number of cases that will not be detected, SVM must be used. For more confident detection of diabetes, it would be better to opt for KNN algorithm. In order to create a good clinical decision support system, it would be better to choose Random Forest algorithms. It should be noted once again that multiple evaluation metrics have to be used for analyzing imbalanced healthcare datasets.

5.2 Limitations and Future Work

This research utilized a set of hyperparameter configurations and train-test divides to test the system’s outputs. While the results generated indicate positive outcome possibilities, improvement opportunities remain (e.g., through hyperparameter tuning, probability calibration, threshold adjustment, class weighting, and class imbalance handling with various resampling techniques). Additionally, testing using an independent external data source will provide further evidence of the system’s generalization ability when utilizing it within an actual environment. Future studies will include precision-recall analysis, calibration curves, and clinically adjusted decision thresholds depending upon the intended use case (screening/triaging). Such optimizations would lead to a greater trust factor associated with the system, thus increasing the system’s potential for providing beneficial outcomes through real-world applications.

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